# NIKIL KRISHNA *Undergraduate Student*

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## ABOUT ME

My academic and research interests lie in structural computational biology. My experience encompasses molecular visualization, design, and simulation. I am currently learning about artificial intelligence and machine learning applications for in silico drug design and protein binder design.

## EXPERIENCE

**GATE Biotechnology** 04/2026
- AIR 14
- Score 858

**Capstone Project and Publication** 04/2026
*SASTRA Deemed University*
- Structure-Based Drug Design of Novel Inhibitors Targeting the Thiamine Biosynthesis Pathway in Bacteria
- High throughput **Virtual Screening**
- **Lead Optimization** by R-Group substitution and SAR analysis
- Molecular Dynamics and Analysis
- Preprint available at bioRxiv

**Founder** 01/2026
*Phage*
- phage.netlify.app
- **Molecular Dynamics** as a Service.
- Frontend **web design**
- Production Grade Molecular Dynamics **Automation**
- Backend Scaling, Optimization and Maintenance
- **Agentic Engineering**

**Summer Internship** 06/2025
*IIT Dharwad*
- Development of Early-Stage Computational **Drug Discovery Pipeline** Using **Python**
- Virtual Screening, Rescoring and Statistical Analysis **Automation**
- Virtual Environment and dependency management using **Conda**
- Parallel High Performance **Scientific Computing**
- Benchmark of three SOTA **machine learning** architectures for protein binder design

**Undergraduate Student Researher** 06/2024
*SASTRA Deemed University*
- Molecular Motors Laboratory
- Drug Design and Protein Engineering
- Research Credits - Fragment Based Drug Design Of Anthelmintic using Genetic Algorithm

## EDUCATION

**B.tech Biotechnology** 2022
*SASTRA Deemed University, Thanjavur*

**HSC and SSLC** 2020
*ALPHA Wisdom Vidyashram*

## SKILLS

**ML Protein Structure Prediction**
- Alphafold
- Boltz
- Diffusion Based Models

**Drug Design**
- Virtual Screening
- Fragment Based Drug Design
- Genetic algorithm/Combinatorial Chemistry
- Lead Optimization
- ADMET analysis

**Molecular Docking and Dynamics**
- Autodock
- Diffdock
- Gromacs
- OpenMM

**Process Simulation**
- Aspen Plus
- SuperPro
- Designer
- DWSIM

**Molecular Modelling**
- Chimerax
- Pymol
- VMD
- MOE
- Schrodinger Maestro

**Python**
- Molecular Visualization
- Data Analysis
- Machine Learning
- Bioinformatics Pipelines
- Dependency management
- Al Automation

**Bioinformatics**
- Linux
- BLAST, MSA
- Visual Studio Code
- Git version management
- Scientific Computing

## LANGUAGES

**English**
Professional working proficiency

**Hindi**
Limited working proficiency

**Tamil**
Bilingual proficiency

**Kannada**
Bilingual proficiency

**Telugu**
Elementary proficiency

## CERTIFICATIONS

**NPTEL**
Aspen Plus Simulation Software - A Basic Course For Beginners
- 90%

**Schrodinger Drug Discovery Hackathon Winner**
Insilco Structure-based Drug Design targeting *Mycobacterium Tuberculosis* using genetic algorithm and combinatorial chemistry.

**NPTEL**
2024 Drug Delivery: Principles and Engineering
- 83%